Activity - 1

Name : K. chandana
Regno : 192525363

① FCFS - Average Waiting Time

Given,

process	Burst time
P1	5ms
P2	3ms
P3	8ms

Formula,

$$WT = CT - AT - BT$$

All processes arive at time 0. so, waiting time is the sum of Burst times of previous processes.

P1 = 0ms.
P2 = 5ms
P3 = 5+3
= 8ms

Averag waiting time formula,

$$AWT = \frac{\Sigma WT}{n}$$

$$= \frac{0+5+8}{3}$$

$$= 13/3$$

AWT = 4.33ms //

② Round cabin - Which process completes first?

Given, Time Quantum = 2ms

process
P1
P2
P3

Rule: each time

Execution:
P1 exec
P2 exe
P3 ex

Formula:

Here, R

For

③ Give

Proc
P
P
P

process	Burst time
P1	5ms
P2	3ms
P3	1ms

Rule: Each process gets cpu time up to Quantum time

Execution:

P1 Executes from 0-2ms; remaining = 3ms

P2 executes from 2-4ms; remaining = 1ms

P3 executes from 4-5ms, remaining = 0ms

Formula: $RT = BT - ET$

Here, RT = Remaining Time, BT = Burst Time, ET = Executed time

For P3,

$RT = 1 - 1$

$\boxed{RT = 0}$

(3) FCFS- Average Turnaround time

Given,

process	Arrival Time	Burst time
P1	0	5ms
P2	2	3ms
P3	4	2ms

Formula 1 : completion time

CT = Time at which process finish

Execution order,

P1 → P2 → P3

P1 : CT = 5

P2 : CT = 5+3 = 8

P3 : CT = 8+2 = 10

Formula 2:

Turn around Time

TAT = CT - AT

process	CT	AT	TAT
P1	5	0	5ms
P2	8	2	6 ms
P3	10	4	6 ms

Formula 3:

Average Turnaround time

$$ATAT = \frac{\Sigma TAT}{n}$$

$$= \frac{5+6+6}{3}$$

$$= 17/3$$

$$= 5.67ms$$

ATAT = 5.67ms

(4) NON-preem

Given,

process

P1

P2

P3

P4

Rule: The

time

by t

Execution

Formula:

WT

All arr

process

P3

P2

P4

P1

Av

④ NON-preemptive SJF - Average waiting time.

Given,

process	Burst time
P1	7ms
P2	4ms
P3	1ms
P4	4ms

Rule: The process with the shortest burst time executes first. Ties are resolved by the process order.

Execution order: P3 → P2 → P4 → P1

Formula:

$$WT = CT - AT - BT$$

All arrival times are 0

process	Start time	Burst time	waiting time
P3	0	1	0ms
P2	1	4	1ms
P4	5	4	5ms
P1	9	7	9ms

Average waiting time ⇒ $AWT = \frac{\Sigma WT}{n}$

$$AWT = \frac{0+1+5+9}{4}$$

$$= 15/4$$

$$= 3.75ms$$

⑤ Non-preempive priority scheduling — Average waiting time

Given, process	Burst time	priority
P1	5ms	3
P2	3ms	1
P3	4ms	2

Assumption:
Smaller priority number means higher priority

Execution order: P2 → P3 → P1

Formula;

$$WT = CT - AT - BT$$

All arrival times or 0.

process	start time	Burst time	waiting time
P2	0	3	0ms
P3	3	4	3ms
P1	7	5	7ms

Average waiting time;

$$AWT = \frac{\Sigma WT}{n}$$

$$= \frac{0+3+7}{3}$$

$$= 10/3$$

$$AWT = 3.33ms$$

⑥ Round

Given,

Time

process

P1
P2
P3

Executi

Time

0-2
2-4
4-5
5-7
7-8
8-9

Formul

C

P2

So,

⑦ R

Pr

⑥ Round Robin - completion time of P2.

Given,

Time Quantum = 2ms

process	Burst time
P1	5ms
P2	3ms
P3	1ms

Execution order:

Time	process	Remaining Burst time
	P1	3ms
0-2	P2	1ms
2-4	P3	0ms
4-5	P1	1ms
5-7	P2	0ms
7-8	P1	0ms
8-9		

Formula:

CT = Time at which process finish

P2 finishes at time 8ms

So, completion time of P2 = 8ms //

⑦ Round Robin - completion times

Given : Time Quantum = 3ms

process	Burst time
P1	6ms
P2	5ms
P3	2ms

Execution:

Time	process	Remaining Burst time
0-3	P1	1ms
3-6	P2	2ms
6-8	P3	0ms
8-9	P1	0ms
9-11	P2	0ms

Formula:

CT = Time at which process finish

∴ Therefore,

P1: CT = 9ms

P2: CT = 11ms

P2: CT = 8ms

P1 = 9ms, P2 = 11ms, P3 = 8ms //